1

Introduction

Pipeline hydraulics deals with the flow of fluids in pipelines. Fluids are defined as liquids and gases. Specifically, this book deals with liquid flow in pipelines. Liquids are considered to be incompressible for most purposes. Today, thousands of miles of pipelines are used to transport crude oil and petroleum products such as gasoline and diesel from refineries to storage tanks and delivery terminals. Similarly, thousands of miles of concrete and steel pipelines are used to transport water from reservoirs to distribution locations. Engineers are interested in the effect of pipe size, liquid properties, pipe length, etc., in determining the pressure required and horsepower necessary for transporting a liquid from point A to point B in a pipeline. It is clear that the heavier the liquid, the more pressure and hence more horsepower required to transport a given quantity for a specified distance. In all these cases we are interested in determining the optimum pipe size required to transport given volumes of liquids economically and safely through the pipelines.

This book consists of 12 chapters that cover the practical aspects of liquid pipeline hydraulics and the economics of pipelines used to transport liquids under steady-state conditions, except Chapter 11, which introduces the reader to unsteady flow. For a more detailed analysis and study of unsteady flow and transients, the reader should consult one of the books listed in the References section at the end of the book. Appendices containing tables and charts, answers to selected problems, and a summary of formulas are also included at the end of this book.

Chapter 2 covers units of measurement, and properties of liquids such as density, gravity, and viscosity that are important in liquid pipeline hydraulics. Chapter 3 discusses pressure, velocity, Reynolds number, friction factor, and pressure drop calculations using various formulas. Several example problems are discussed and solved to illustrate the various methods currently used in pipeline engineering.

Chapter 4 is devoted to the strength analysis of pipes. It addresses allowable internal working pressures and hydrostatic test pressures and how they are calculated.

Chapter 5 extends the concepts developed in Chapter 3 by analyzing the total pressure and horsepower required to pump a liquid through long-distance pipelines with multiple pump stations, including the transportation of high vapor pressure liquids, such as liquefied petroleum gas (LPG). Injection and delivery along a long pipeline and branch pipe analysis are also covered, and the use of pipe loops to reduce friction and increase throughput is analyzed.

Chapter 6 deals with optimizing pump station locations in a trunk line, minimizing pipe wall thickness using telescoping and pipe grade tapering. Open channel flow, slack line operation in hilly terrain, and batching different products are also addressed in this chapter.

Chapter 7 covers centrifugal pumps and positive displacement pumps applied to pipeline transportation. Centrifugal pump performance curves, Affinity Laws, and the effect of viscosity are discussed as well as the importance of net positive suction head (NPSH). Operation of pumps in series and parallel, and modifications needed to operate pumps effectively, are also covered in this chapter.

Chapter 8 discusses pump station design, minimizing energy loss due to pump throttling with constant-speed motor-driven pumps. The advantages of using variable speed drive (VSD) pumps are also explained and illustrated with examples.

Chapter 9 introduces the reader to thermal hydraulics, pressure drop calculations, and temperature profiles in a buried heated liquid pipeline. The importance of thermal conductivity, overall heat transfer coefficient, and how they affect heat loss to the surrounding are covered.

Chapter 10 introduces flow measurement devices used in measuring liquid flow rate in pipelines. Several of the more common instruments such as the Venturi meter, flow nozzle, and orifice meter are discussed and calculation methods explained.

Chapter 11 gives a basic introduction to unsteady flow and transient hydraulic analysis. This is an advanced concept that would require a separate book to cover the subject fully. Therefore, this chapter will serve as a starting point in understanding transient pipeline hydraulics. The reader should consult one of the publications listed in the References section for a more detailed study of unsteady flow and pipeline transients.

Chapter 12 addresses economic aspects related to pipeline feasibility studies. In addition, the pipeline and pump station capital cost, annual operating cost, and calculation of transportation tariff are discussed. Also covered in this chapter is the analysis of the optimum pipe size and pumping equipment required that produces the least cost. A discounted cash flow approach using the present value (PV) of investment is employed in determining the optimum pipe size for a particular application.

In each chapter, example problems are used to illustrate the concepts introduced. Problems for practice are also included at the end of each chapter. Answers to selected problems may be found in Appendix B.

Appendix A consists of tables and charts containing units and conversions, common properties of petroleum fluids, etc.

In addition, for quick reference, formulas used in all chapters have been assembled and summarized in Appendix C.